

Midterm Exam: Cell & Tissue Culture Techniques

Section I: Multiple Choice Questions (MCQs)

1. The term 'In Vitro' literally means:
 - a) Inside the body
 - b) In glass
 - c) In nature
 - d) Growing organs
2. Which of the following is used to achieve a sterile environment in the lab?
 - a) Centrifuge
 - b) Water bath
 - c) Laminar flow cabinet
 - d) Inverted microscope
3. The ideal temperature for culturing mammalian cells is:
 - a) 25°C
 - b) 4°C
 - c) 37°C
 - d) -20°C
4. Which enzyme is used to detach adherent cells from the culture flask?
 - a) Amylase
 - b) Trypsin
 - c) Glutamine
 - d) Insulin
5. HeLa cells are an example of:
 - a) Primary cells
 - b) Finite cell lines
 - c) Continuous cell lines
 - d) Blood cells
6. Cryopreservation of cell stocks is typically done in:
 - a) CO2 Incubator
 - b) Liquid nitrogen (-196°C)
 - c) Refrigerator (4°C)
 - d) Autoclave
7. Cells that must attach to a solid surface to grow are called:
 - a) Suspension cells



- b) Primary cells
 - c) Adherent cells
 - d) Plant cells
8. The CO₂ concentration in an incubator (usually 5%) is essential for:
- a) Providing energy
 - b) Maintaining pH
 - c) Killing bacteria
 - d) Freezing cells
9. Which of the following is NOT a component of cell culture media?
- a) Amino acids
 - b) Glucose
 - c) Serum
 - d) HEPA filter
10. The process of transferring cells to a new vessel with fresh media is called:
- a) Passaging (Subculturing)
 - b) Sterilization
 - c) Digestion
 - d) Incubation

Section II: True or False

1. Primary cells have a limited lifespan known as the Hayflick limit. ()
2. An inverted microscope has the light source below and objectives above. ()
3. HEPA filters are used in laminar flow cabinets to filter air and remove contaminants. ()
4. Continuous cell lines are usually derived from normal, healthy tissues. ()
5. Secondary cell culture is made directly from living tissue using enzymatic digestion. ()
6. Serum (e.g., FBS) provides growth factors and hormones for cell proliferation. ()
7. Autoclaves are used for sterilization work in the laboratory. ()
8. Suspension cells require enzymes like Trypsin to be detached. ()
9. High humidity (~95%) in the incubator prevents the evaporation of the media. ()
10. Synthetic media was widely used starting from the early 1900s. ()

Section III: Complete the following sentences

1. _____ cells grow in a single layer on the bottom of the flask.



2. The _____ is the first animal cell culture grown by Harrison in 1907.
3. For long-term preservation, cells are stored in _____ containers.
4. _____ cells can divide indefinitely.
5. The main concern in cell culture that must be prevented is _____.
6. Medium components like glutamine and serum are kept at _____ °C.
7. _____ microscope is used for monitoring the cells during growth.
8. A _____ is used for pelleting cells during the subculturing process.
9. _____ is the latin term meaning 'outside the living organism'.
10. To prevent contamination, you should spray gloves with _____% ethanol.



Answer Key

Section I: MCQs

1-b, 2-c, 3-c, 4-b, 5-c, 6-b, 7-c, 8-b, 9-d, 10-a

Section II: True or False

1-True, 2-False (Light source above, objectives below), 3-True, 4-False (usually from tumors), 5-False (from primary culture), 6-True, 7-True, 8-False (adherent cells do), 9-True, 10-False (1950s-60s)

Section III: Completion

1-Adherent, 2-Primary, 3-Cryostorage (Liquid nitrogen), 4-Continuous (Immortalized), 5-Contamination, 6- -20°C, 7-Inverted, 8-Centrifuge, 9-In vitro, 10- 70%

